

OKUNEV NOS, Russian Federation

WMO: 233050

Lat: 66.2574N

Lon: 52.5642E

Elev: 22

StdP: 101.06

Time Zone: 3.00 (E03)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.1	(c) -33.5	(d) -39.9	(e) 0.1	(f) -36.9	(g) -36.3	(h) 0.1	(i) -33.3	(j) 8.4	(k) -7.4	(l) 7.6	(m) -7.7	(n) 1.1	(o) 0	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.3	(c) 26.7	(d) 17.4	(e) 24.4	(f) 16.6	(g) 22.3	(h) 15.7	(i) 18.5	(j) 24.0	(k) 17.6	(l) 22.7	(m) 16.6	(n) 21.1	(o) 3.2	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.6	11.8	20.5	15.6	11.1	19.6	14.6	10.4	19.3	52.4	24.2	49.6	22.7	46.9	21.0	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.1	(b) 7.1	(c) 6.4	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 8.1	(b) 7.1	(c) 6.4	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.6	-18.2	-11.8	-8.7	-1.7	5.0	11.2	15.7	12.6	7.3	-0.3	-7.3	-11.5		
	DBStd	12.38	9.87	8.37	8.02	4.64	5.81	4.96	4.93	3.67	3.42	4.26	6.93	8.39		
	HDD10.0	4259	874	611	580	351	181	44	8	14	95	319	518	666		
	HDD18.3	6946	1132	844	838	601	415	219	113	182	332	577	768	924		
	CDD10.0	397	0	0	0	0	25	80	184	94	13	0	0	0		
	CDD18.3	41	0	0	0	0	1	6	31	4	0	0	0	0		
	CDH23.3	318	0	0	0	0	12	40	239	27	0	0	0	0		
	CDH26.7	66	0	0	0	0	1	6	56	3	0	0	0	0		
(14) Wind	WSAvg	3.3	3.0	3.5	3.5	3.3	3.5	3.5	3.0	3.1	3.0	3.2	3.2	3.2		
(15) Precipitation	PrecAvg	511	35	26	27	31	40	60	62	68	61	53	38	35		
	PrecMax	595	50	39	53	62	76	137	143	118	237	91	65	49		
	PrecMin	425	16	13	8	14	20	25	24	34	23	20	15	17		
	PrecStd	50	9	8	13	15	15	29	29	19	49	17	14	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.9	2.1	5.3	12.4	25.1	27.1	30.9	26.2	19.1	8.5	3.2	1.6		
		MCWB	0.3	1.3	2.3	6.6	14.7	17.9	18.2	18.4	15.0	7.1	2.1	0.9		
	2%	DB	-0.7	0.9	2.7	9.4	20.8	23.9	28.2	23.0	16.0	7.1	2.0	0.7		
		MCWB	-1.2	0.1	0.6	4.5	12.9	15.8	17.6	16.9	12.6	5.9	1.2	0.2		
	5%	DB	-2.6	-0.2	1.7	6.8	17.3	21.9	26.0	20.8	13.7	6.0	1.2	-0.4		
		MCWB	-3.1	-1.0	0.2	3.1	10.9	14.8	17.0	15.9	11.1	5.2	0.6	-1.0		
	10%	DB	-4.5	-1.7	0.9	4.7	14.6	19.5	23.9	18.8	12.2	4.9	0.5	-1.4		
		MCWB	-5.0	-2.4	-0.3	2.5	9.4	14.0	16.6	14.6	10.5	4.1	0.0	-2.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.4	1.2	2.3	6.6	15.1	19.0	20.1	19.3	15.1	7.4	2.3	1.1		
		MCDB	0.7	1.6	4.4	11.4	22.9	24.6	26.5	24.2	18.2	8.1	2.8	1.3		
	2%	WB	-1.1	0.2	1.1	4.7	13.3	17.5	19.1	17.7	13.0	6.2	1.3	0.4		
		MCDB	-0.7	0.8	2.2	7.5	19.9	22.3	25.0	21.6	15.4	6.8	1.8	0.8		
	5%	WB	-3.1	-1.0	0.5	3.7	11.5	16.1	18.0	16.6	11.6	5.1	0.7	-0.9		
		MCDB	-2.7	-0.4	1.6	6.0	16.3	19.6	23.7	20.0	13.2	5.9	1.1	-0.5		
	10%	WB	-5.0	-2.4	-0.3	2.8	9.7	14.5	17.2	15.3	10.6	4.2	0.1	-1.9		
		MCDB	-4.5	-1.7	0.9	4.6	14.2	18.8	22.7	18.3	12.1	4.9	0.5	-1.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	6.0	7.9	7.5	8.2	8.0	8.3	7.3	5.4	3.7	4.9	5.5		
		MCDBR	6.6	4.2	4.9	9.6	12.8	11.6	11.6	10.3	8.8	3.8	4.2	4.3		
		MCWBR	6.4	4.4	3.6	6.1	6.7	5.5	4.1	5.8	5.5	3.2	4.2	4.3		
		MCDBR	6.3	4.5	4.1	7.4	12.1	10.1	10.0	9.0	7.2	3.5	4.2	4.0		
	5% WB	MCWBR	6.2	4.6	3.4	5.1	6.9	6.2	4.4	5.7	5.1	3.0	4.2	4.0		
(40) Clear-Sky Solar Irradiance	taub		0.215	0.265	0.303	0.335	0.345	0.352	0.405	0.368	0.328	0.288	0.213	0.214		
	taud		1.910	2.053	2.096	2.134	2.370	2.466	2.312	2.401	2.487	2.282	1.953	1.922		
	Ebn at Noon		189	569	740	815	856	858	788	785	741	557	182	17		
	Edh at Noon		18	57	91	114	101	95	108	89	65	46	15	4		
(44) All-Sky Solar Radiation	RadAvg		0.09	0.69	2.22	3.86	4.93	5.04	4.88	3.22	1.72	0.58	0.14	0.01		
	RadStd		0.01	0.07	0.22	0.38	0.48	0.58	0.61	0.34	0.17	0.08	0.02	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	-406	-448	N/A	N/A

Nomenclature: See separate page